

Lab 17 - Frequency Domain Filtering

June 7, 2025

1 Theory

Frequency domain filtering is a technique in image processing where operations are performed on the Fourier transform of an image rather than directly on the spatial domain. The main steps are:

1. **Fourier Transform:** The image $f(x, y)$ is transformed into its frequency representation $F(u, v)$ using the 2D Discrete Fourier Transform (DFT):

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

2. **Filtering:** A filter function $H(u, v)$ (such as low-pass or high-pass) is applied in the frequency domain:

$$G(u, v) = H(u, v) \cdot F(u, v)$$

- **Low-pass filters** allow low frequencies (smooth regions) and attenuate high frequencies (edges, noise).
 - **High-pass filters** allow high frequencies (edges, details) and attenuate low frequencies.
3. **Inverse Fourier Transform:** The filtered image is obtained by applying the inverse DFT:

$$g(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} G(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

This process enhances or suppresses specific features in the image based on their frequency content.

```
[1]: from scipy import fftpack
import numpy as np
import imageio
from PIL import Image, ImageDraw
import matplotlib.pyplot as plt
```

```
[2]: def frequency_image(image):
    image_array = np.array(image)
    fft1 = fftpack.fftshift(fftpack.fft2(image_array))
    magnitude_spectrum = 20*np.log(np.abs(fft1))
    freq_image = Image.fromarray(magnitude_spectrum)
    freq_image = freq_image.convert("L")
    return freq_image
```

```
[3]: def low_pass_filter(image1):
    #convert image to numpy array
    image1_np=np.array(image1)
    #fft of image
    fft1 = fftpack.fftshift(fftpack.fft2(image1_np))
    #Create a low pass filter image
    x,y = image1_np.shape[0],image1_np.shape[1]

    #defining filter
    #size of circle
    e_x,e_y=50,50
    #create a box
    bbox=((x/2)-(e_x/2),(y/2)-(e_y/2),(x/2)+(e_x/2),(y/2)+(e_y/2))
    low_pass=Image.new("L",(image1_np.shape[0],image1_np.shape[1]),color=0)
    draw1=ImageDraw.Draw(low_pass)
    draw1.ellipse(bbox, fill=1)
    low_pass_np=np.array(low_pass)
    low_pass_np = low_pass_np.T
    #end of defining filter

    #multiply both the images
    filtered=np.multiply(fft1,low_pass_np)

    #inverse fft
    ifft2 = np.real(fftpack.ifft2(fftpack.ifftshift(filtered)))
    ifft2 = np.maximum(0, np.minimum(ifft2, 255))
    data = Image.fromarray(ifft2)
    data = data.convert("L")

    return data
```

```
[4]: def high_pass_filter(image):
    #converting image to array
    image_array = np.array(image)

    #sending image to low pass filter
    lowpass_image = low_pass_filter(image)
    #converting image to array
    lowpass_image_array = np.array(lowpass_image)

    #subtracting lowpass image from original to obtain highpass image
    high_pass_array = image_array - lowpass_image_array

    #array to image
    high_pass_image = Image.fromarray(high_pass_array)
```

```

high_pass_image = high_pass_image.convert("L")

return high_pass_image

```

```

[ ]: image = imageio.imread('img/samir.jpg', mode='F')
freq_image = frequency_image(image)
lowpass_image = low_pass_filter(image)
highpass_image = high_pass_filter(image)

fig = plt.figure()
fig.set_figheight(20)
fig.set_figwidth(20)

#plotting original image
fig.add_subplot(2,2,1)
plt.imshow(image, cmap='gray')
plt.title('original')

#plotting the image in frequency domain
fig.add_subplot(2,2,2)
plt.imshow(freq_image, cmap='gray')
plt.title('Frequency Domain')

#plotting lowpass image
fig.add_subplot(2,2,3)
plt.imshow(lowpass_image, cmap='gray')
plt.title('lowpass_image')

#plotting highpass image
fig.add_subplot(2,2,4)
plt.imshow(highpass_image, cmap='gray')
plt.title('highpass_image')

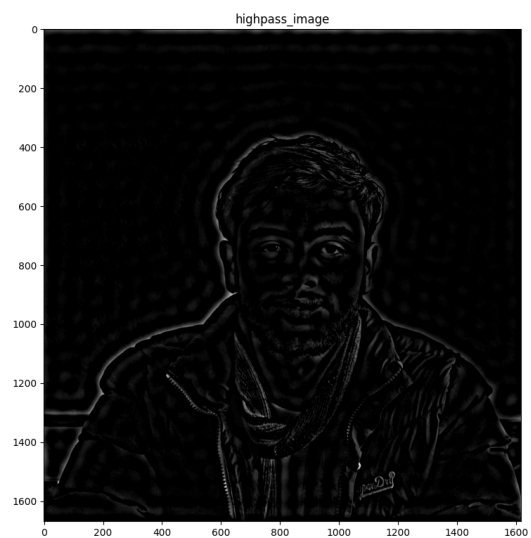
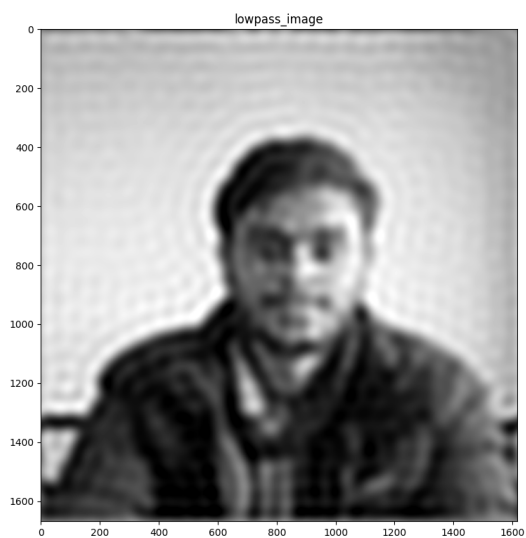
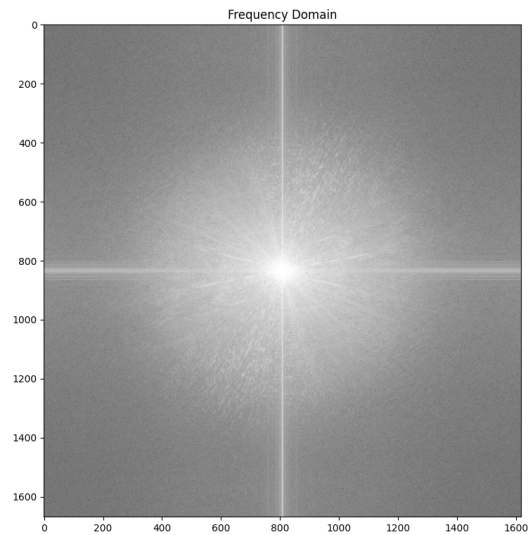
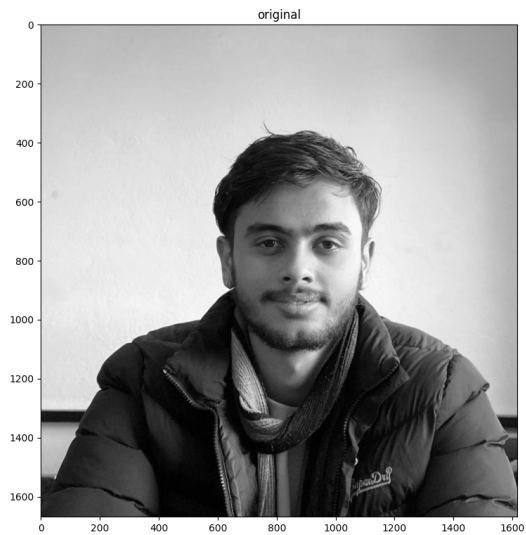
```

C:\Users\samir\AppData\Local\Temp\ipykernel_27756\909931437.py:1:
DeprecationWarning: Starting with ImageIO v3 the behavior of this function will switch to that of iio.v3.imread. To keep the current behavior (and make this warning disappear) use `import imageio.v2 as imageio` or call `imageio.v2.imread` directly.
image = imageio.imread('img/samir.jpg', mode='F')

```

[ ]: Text(0.5, 1.0, 'highpass_image')

```



The Kernel crashed while executing code in the current cell or a previous cell.

Please review the code in the cell(s) to identify a possible cause of the failure.

Click [here](\"https://aka.ms/vscodeJupyterKernelCrash\") for more info.

View Jupyter [log](\"command:jupyter.viewOutput\") for further details.

2 Conclusion

Frequency domain filtering provides powerful tools for image enhancement by manipulating the frequency components. Low-pass filters smooth images and reduce noise, while high-pass filters enhance edges and fine details. By applying these filters in the frequency domain and transforming the result back to the spatial domain, we can achieve effects that are difficult or inefficient to realize with spatial domain techniques alone. This lab demonstrates the practical application of Fourier-based filtering for image analysis and enhancement.